



University of Tehran

School of Electrical and Computer Engineering (ECE)

Large Language Models : CA1

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This assignment was completed as part of the coursework for the Large Language Models class in the Master's program.

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1 Dataset and Tokenizer

According to the outputs that we've gotten(which are present in the code file): The dataset contains 93889 unique names (and 1825433 names in general). The maximum length among all names is 15 characters. We created the needed vocabulary. It consists of 29 tokens: 26 lowercase letters (a-z) plus three special tokens ($\langle \text{SOS} \rangle$, $\langle \text{EOS} \rangle$, $\langle \text{PAD} \rangle$).

After preprocessing, the input tensor has shape (1825433, 20) and the target tensor has shape (1825433, 20). Each sequence is padded to a fixed length of 20 tokens.

For the name "Alice": Input: [1, 3, 14, 11, 5, 7, 0, ..., 0] and Target: [3, 14, 11, 5, 7, 2, 0, ..., 0].

For a sample name "mary"(After lowering), the input sequence is: [1, 15, 3, 20, 27, 0, ..., 0] where 1 = $\langle \text{SOS} \rangle$, 15 = 'm', 3 = 'a', 20 = 'r', 27 = 'y', and 0 = $\langle \text{PAD} \rangle$. The target sequence is: [15, 3, 20, 27, 2, 0, ..., 0] where 2 = $\langle \text{EOS} \rangle$.

2 Positional Encoding

We plotted the positional encoding graph for 20 positions:

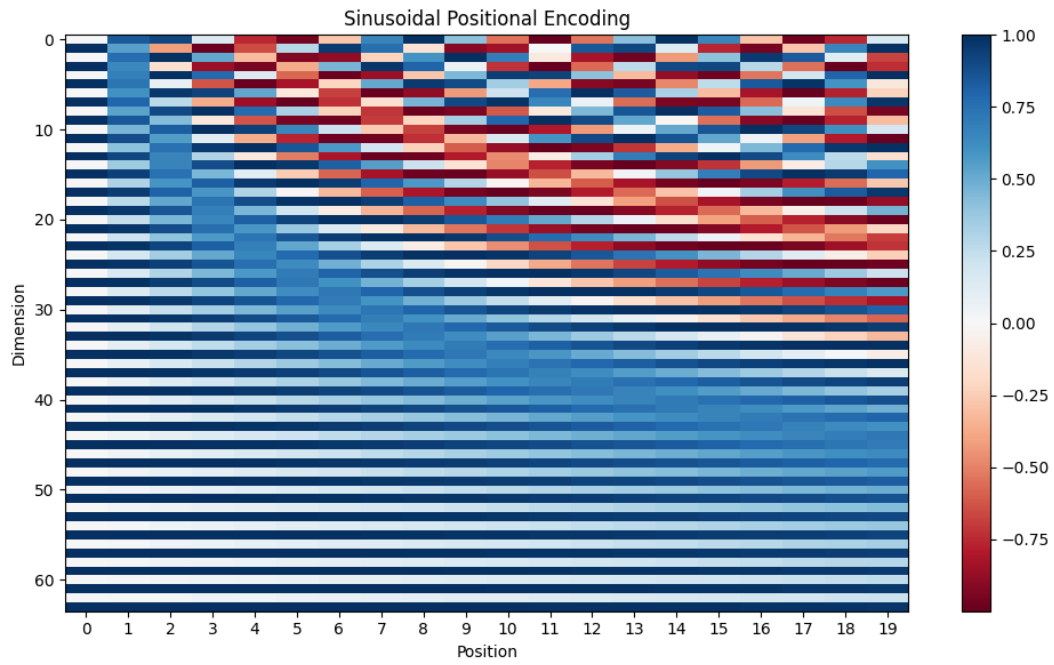


Figure 1: Positional encoding graph

3 Architecture and Training

The decoder-only model that we've built contains a total of 153,821 trainable parameters.

The model was trained on CPU for 3 epochs. The training results are summarized in Table 1.

Table 1: Training results over 3 epochs

Epoch	Time (s)	Train Loss	Val Loss	Device
1	1967.62	2.0360	1.8712	CPU
2	2038.71	1.8908	1.8039	CPU
3	2257.28	1.8481	1.7673	CPU

Both training loss and validation loss decreased consistently across epochs. This indicates that the model is learning meaningful patterns without severe overfitting (both losses decline together).

And here is the desired plot for loss values across epochs:

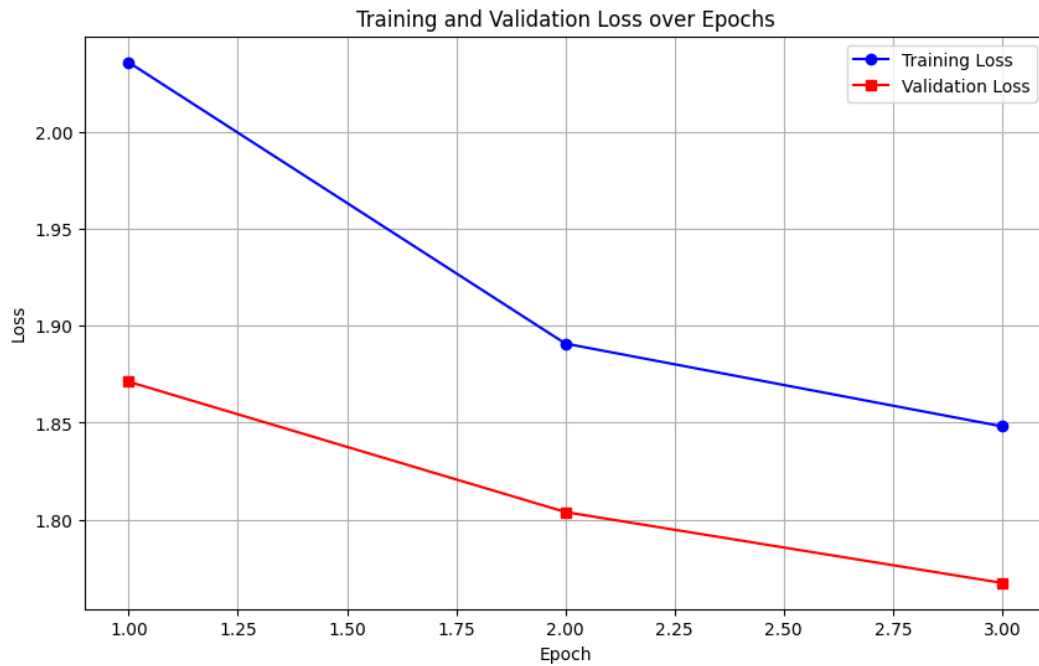


Figure 2: Plot of training and validation loss across epochs

This plot confirms a relatively effective learning without any sign of overfitting!

4 Inference

The generation results for different input prefixes and temperature values are presented in Tables 2-5.

Input: “c”

Temperature	Generated Names
0.0	cad, cad, cad, cad, cad
0.5	claudio, carrol, carmella, caroleen, carmel
1.0	chas, chanestre, cassanton, caishonda, clore-anzo

Table 2: Generated names for input “c”

Input: “li”

Temperature	Generated Names
0.0	lia, lia, lia, lia, lia
0.5	lindsey, lillian, lindsey, lindsey, lillian
1.0	linda, lianna, linesha, lisen, lindey

Table 3: Generated names for input “li”

Input: “mar”

Temperature	Generated Names
0.0	mara, mara, mara, mara, mara
0.5	marian, marian, mariya, marisa, marian
1.0	marcelle, marvelle, maryl, marielda, mariya

Table 4: Generated names for input “mar”

Input: “a”

Temperature	Generated Names
0.0	aba, aba, aba, aba, aba
0.5	alonia, alexandra, alla, alexandra, ashawn
1.0	ayonna, aulysa, adalia, almeth, aliywah

Table 5: Generated names for input “a”

The temperature parameter controls the randomness of token sampling. Higher temperature increases creativity and diversity but may reduce name realism.

Lower temperature produces safer, more repetitive completions. *temperature* = 0 actually equals to Greedy!

So we get some deterministic and repetitive outputs when *temperature* = 0 like “cad”, “lia”, “mara” and ”aba”(All of them were present in the dataset)! In this case, our model selects the character with the most probability! When *temperature* = 0.5, we have a balance between diversity and realism! Some plausible names can be generated like “claudio”, “lindsey”, “marian”, “alexandra”(that were present in the dataset) with some repetition! But some of the generated names were not in the dataset!!! When *temperature* = 1 we are at the highest level of creativity and also diversity! Some unusual names can be produced in this level like “chanestre”, “cassanton”, “aliywah”(not present in the dataset)!